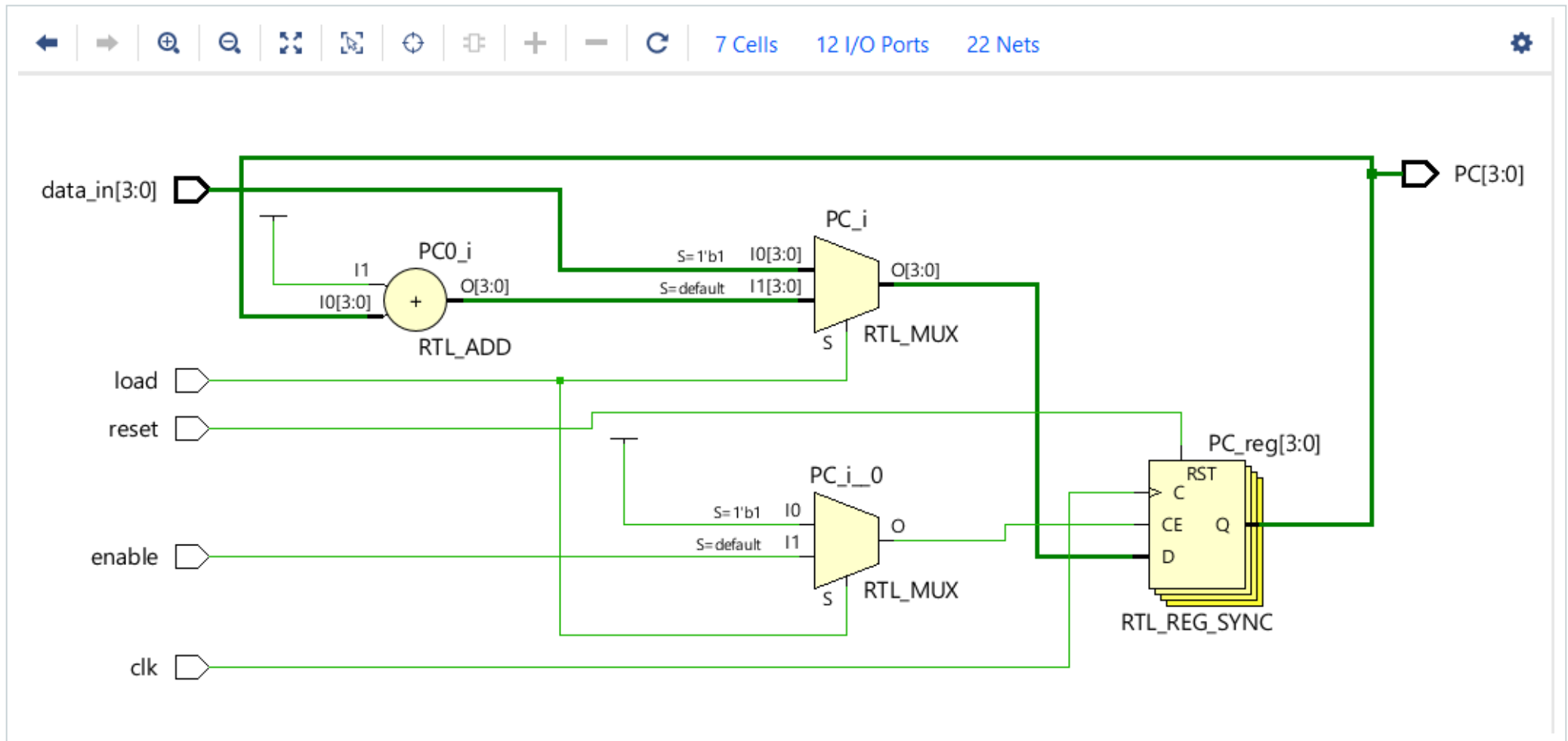
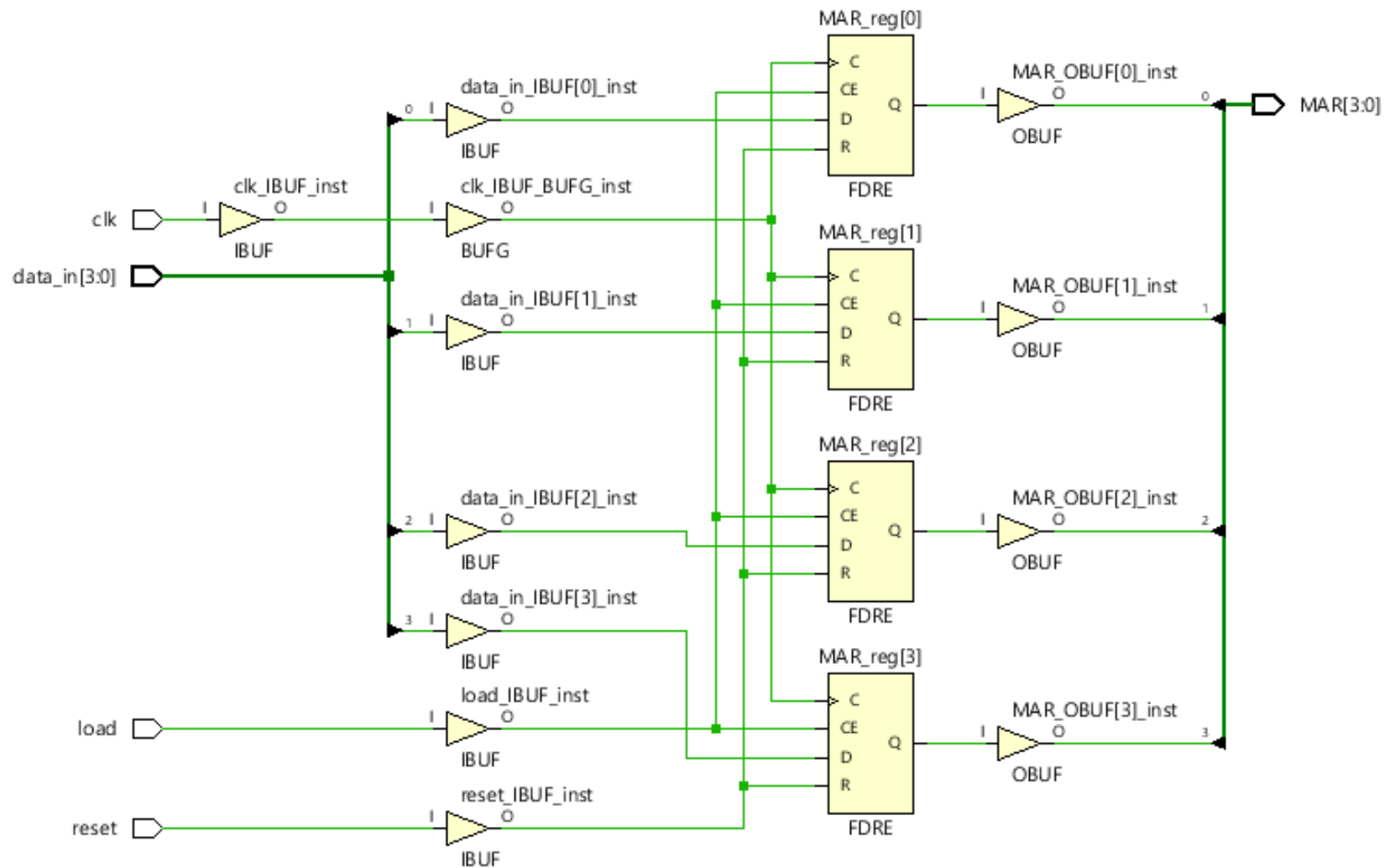


Schematic Design of PC (Program Counter)



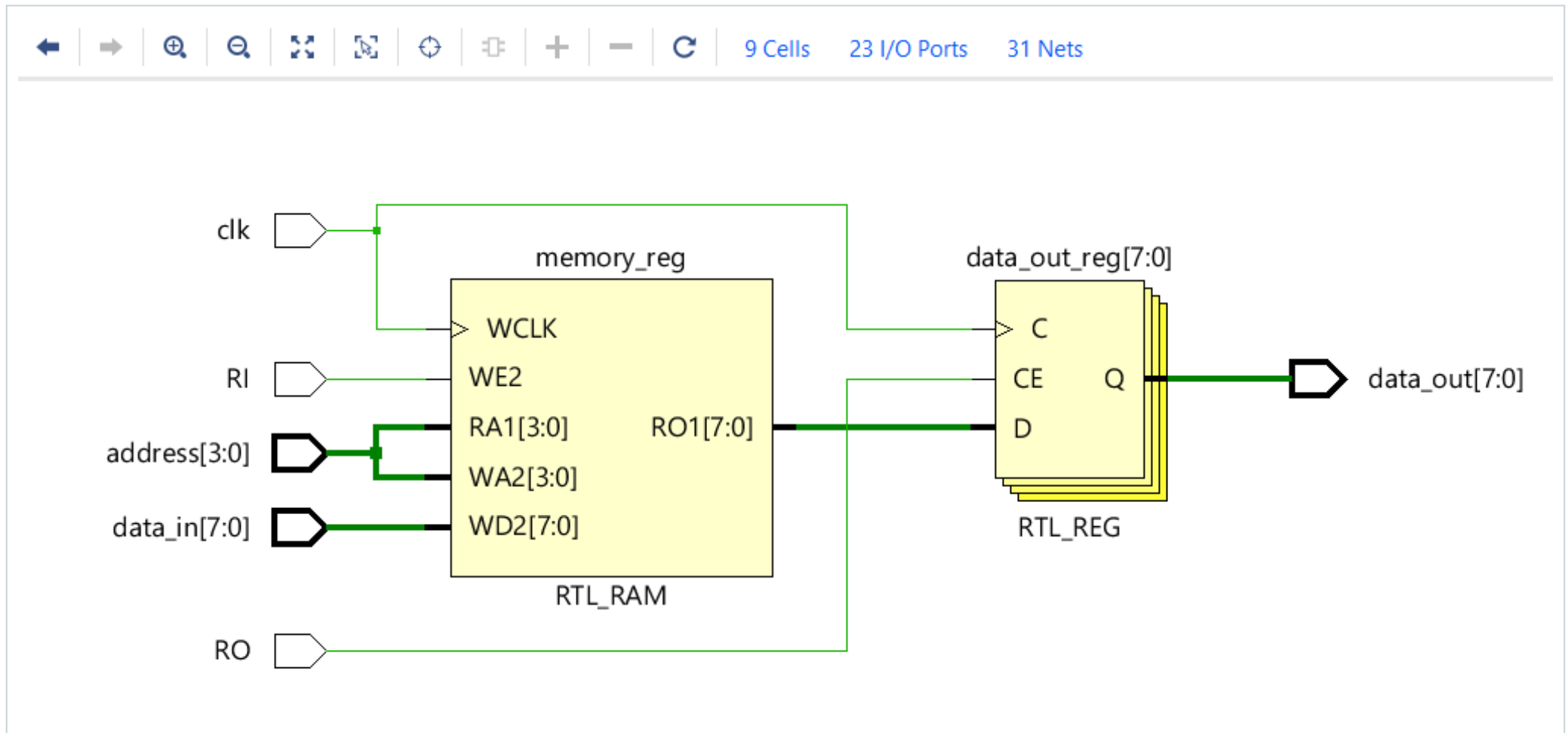
4-bit counter: RTL_ADD increments PC, a mux selects load data or the incremented value, and a synchronous register with enable and reset holds the count.

Schematic Design of MAR (Memory Address Register)



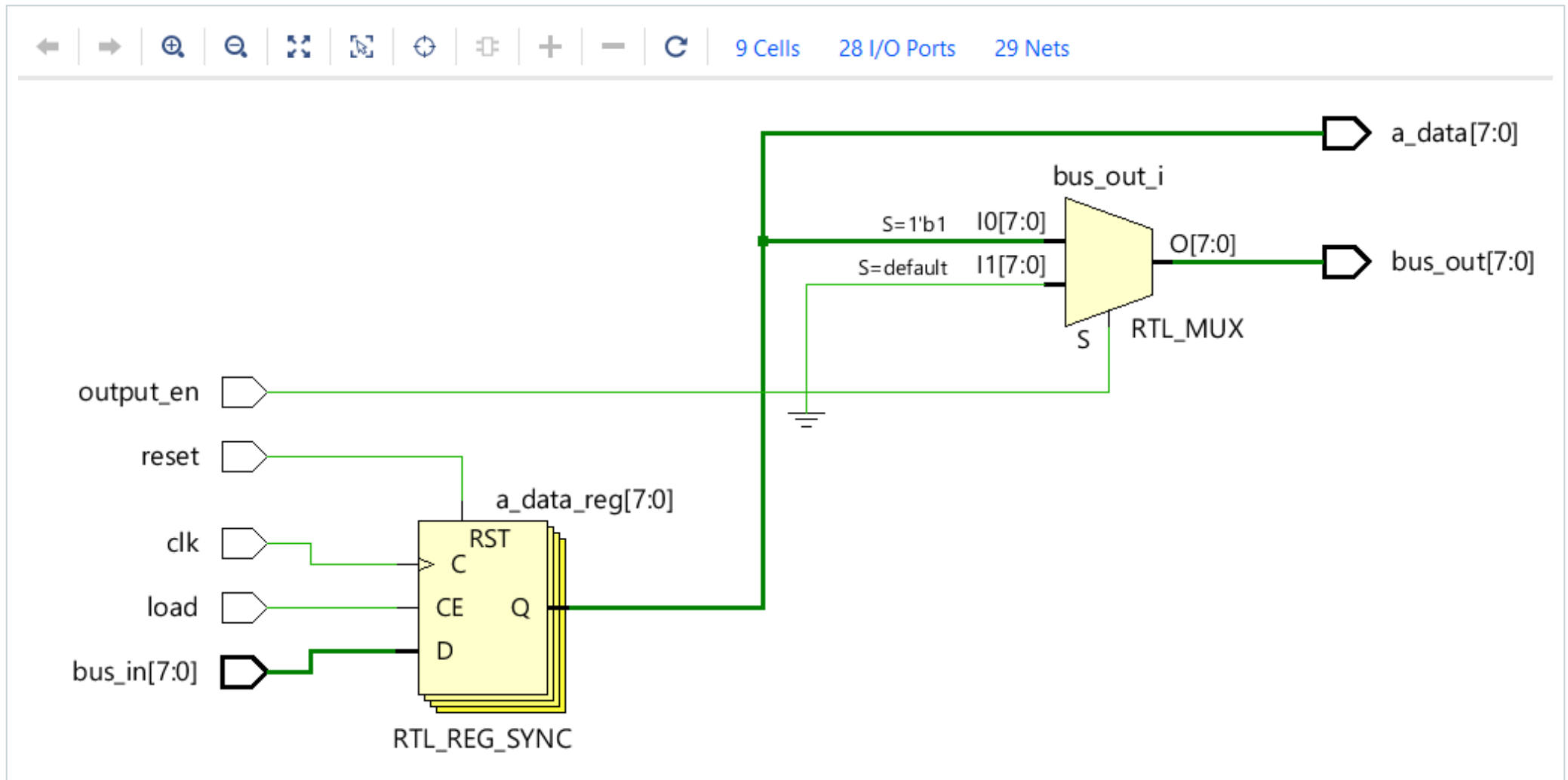
4-bit register built from FDRE flip-flops with load and reset, shown with input and output buffers (synthesized netlist view).

Schematic Design of RAM



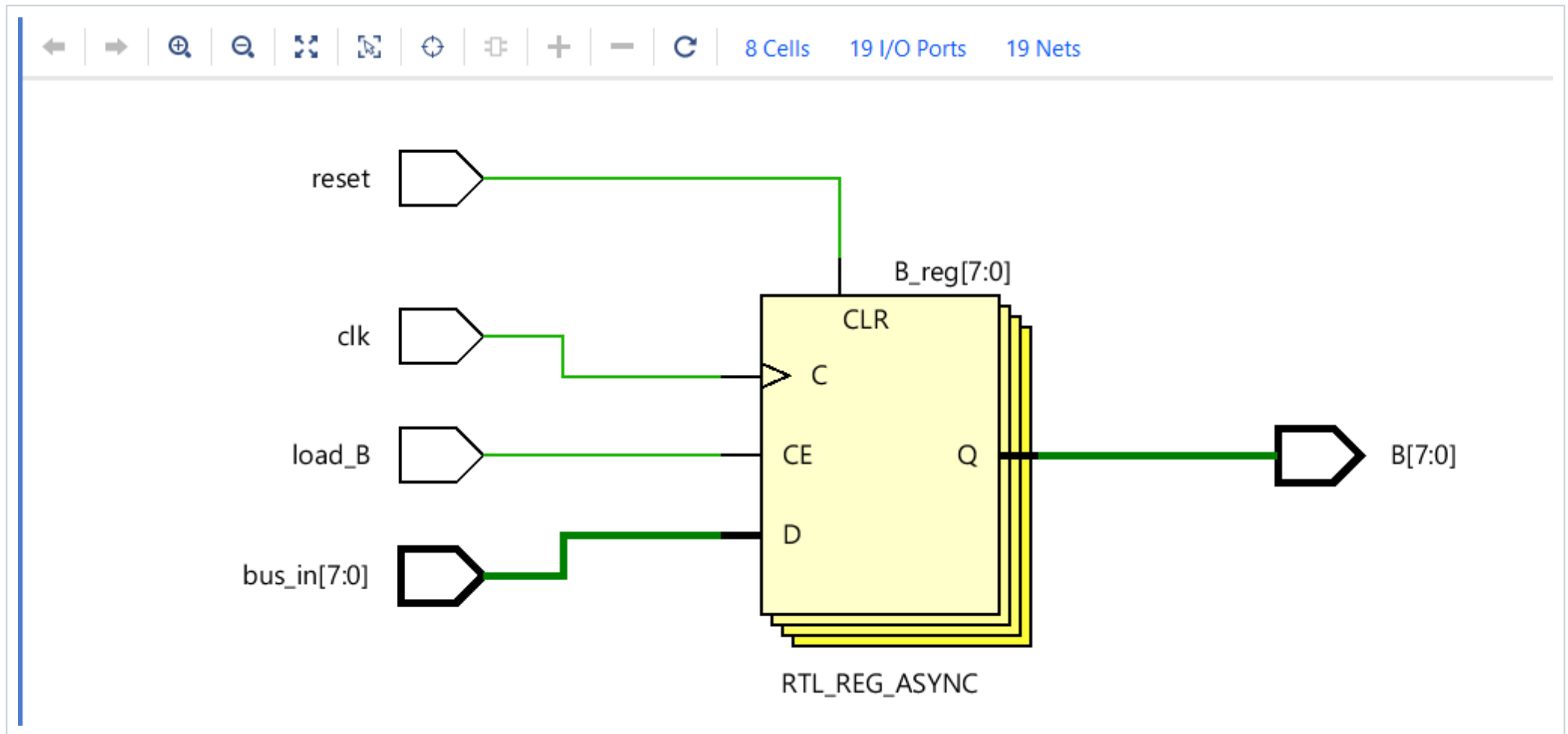
Memory array with a 4-bit address and 8-bit data in, write enable (RI), and an 8-bit output register enabled by RO.

Schematic Design of Register A (Accumulator)



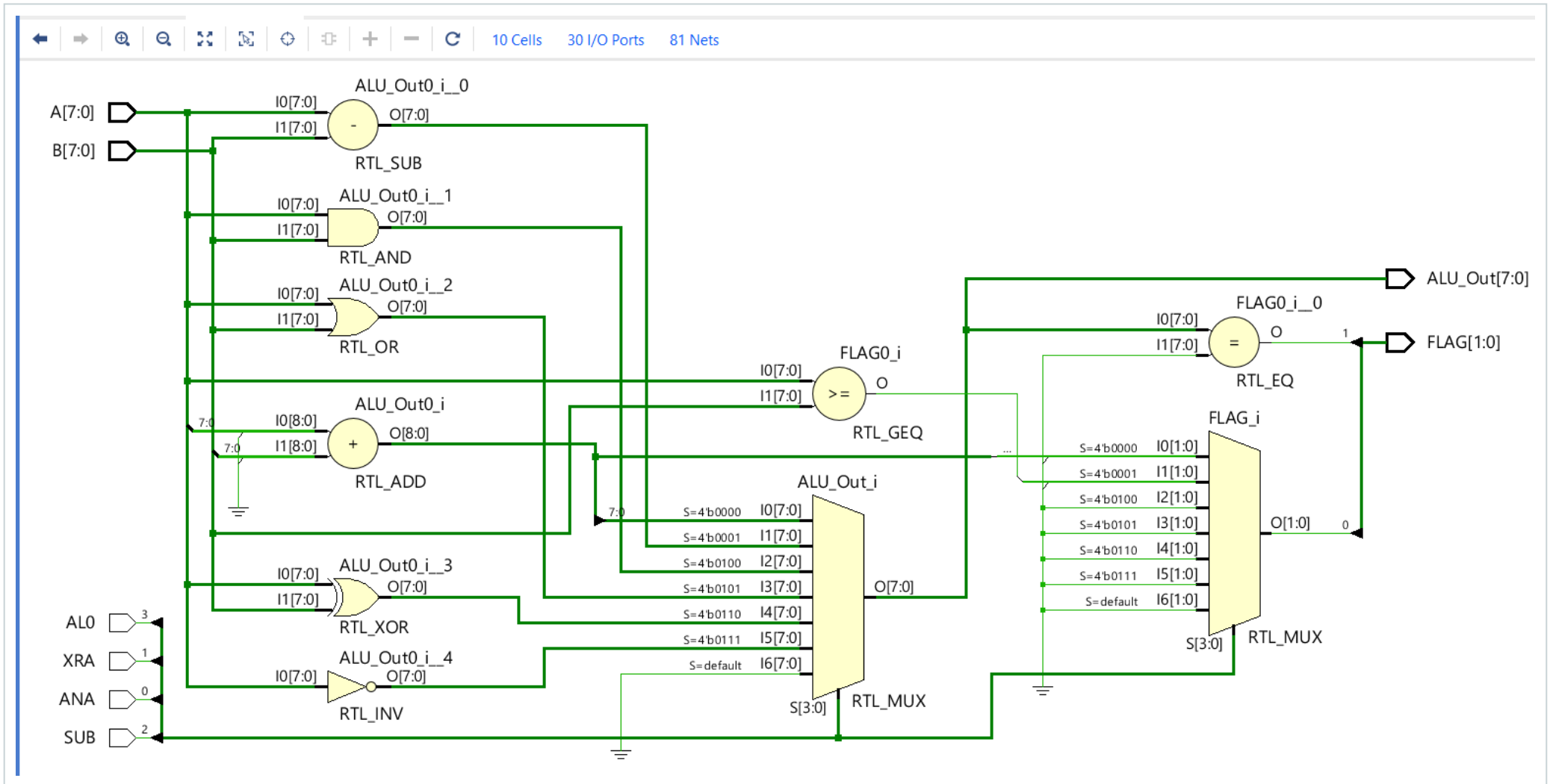
8-bit synchronous register with load and reset; a mux gated by **output_en** drives its value onto **bus_out**.

Schematic Design of Register B



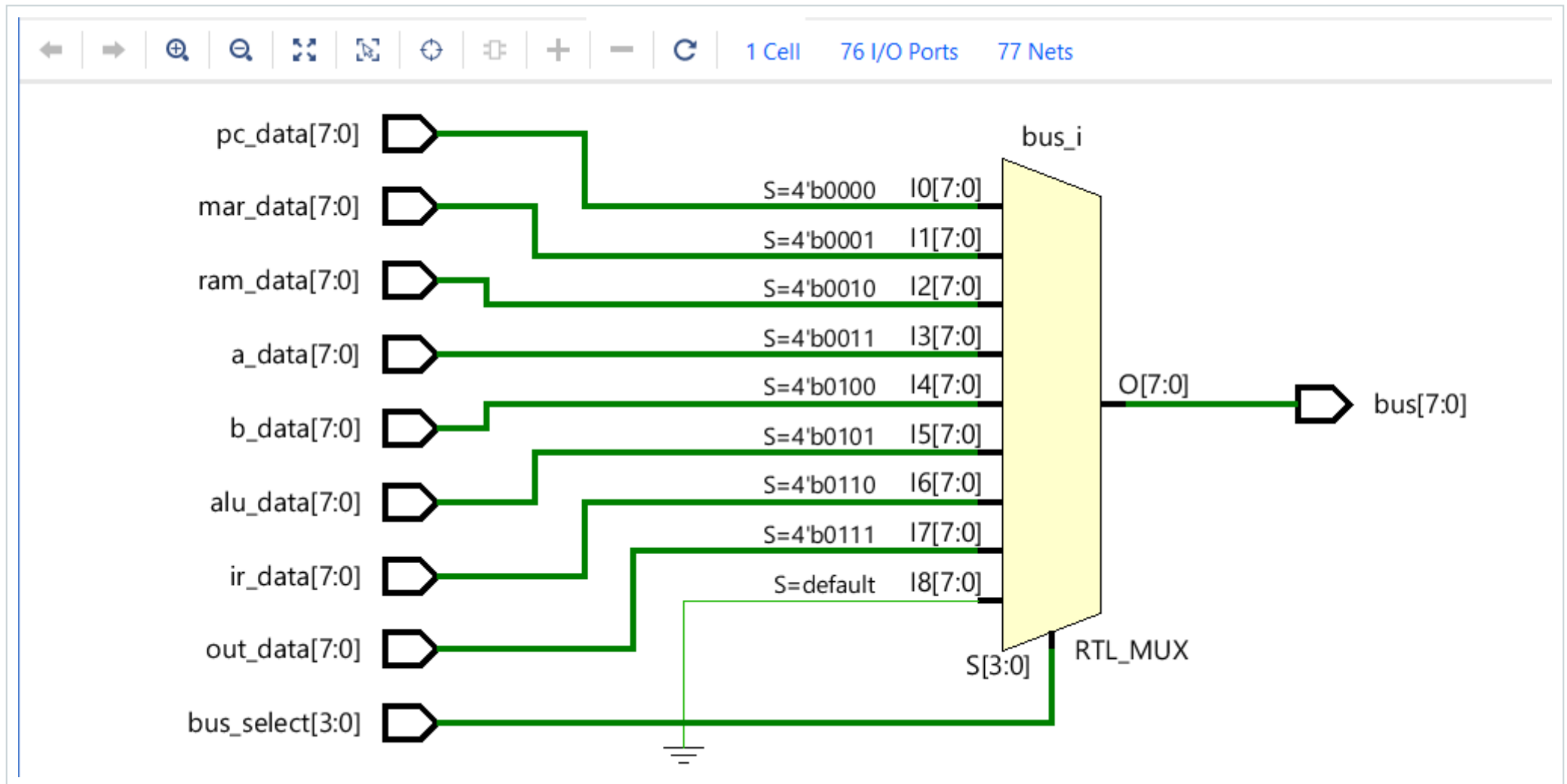
8-bit register loaded from the bus when load_B is asserted, with asynchronous clear on reset.

Schematic Design of ALU8



Subtract, AND, OR, add, XOR and invert units feed a result mux selected by {AL0, SUB, XRA, ANA}; a second mux produces FLAG[1:0].

Schematic Design of Bus Multiplexer



Nine-way mux with a 4-bit bus_select picks PC, MAR, RAM, A, B, ALU, IR or OUT data onto the shared 8-bit bus.